


SLIM - Sparse Linear Methods

$$\frac{1}{2} \|R - \hat{R}\|_F^2 + \lambda \|W\| + \frac{\beta}{2} \|W\|_F^2 \rightarrow \min_W$$

$$\text{s.t. } W_{ij} \geq 0; \quad W_{ii} = 0; \quad \hat{R} = RW$$

for one column:
 $r_j = R w_j$ ← column

$$\frac{1}{2} \sum_j \|r_j - R w_j\|_F^2 + \lambda \sum_j \|w_j\| + \frac{\beta}{2} \sum_j \|w_j\|^2 \rightarrow$$

→ min_W

$$\forall j: \frac{1}{2} \|r_j - R w_j\|^2 + \lambda \|w_j\| + \frac{\beta}{2} \|w_j\|^2 \rightarrow \min_{w_j}$$

$$\mathcal{L}_j = \frac{1}{2} w_j^T R^T R w_j - r_j^T R w_j + \frac{1}{2} r_j^T r_j +$$

$$+ \lambda \|w_j\| + \frac{\beta}{2} \|w_j\|^2$$

$$R w_j = \sum_k r_k w_{kj}$$

$$\frac{\partial \mathcal{L}_j}{\partial w_{kj}} = w_j^T R^T \underline{r_k} - r_k^T r_j + \lambda + \beta w_{kj}$$

$$\frac{\partial \mathcal{L}_j}{\partial w_{kj}} = \sum_i w_{ij} r_i^T r_k - r_k^T r_j + \lambda + \beta w_{kj} \ominus$$

$$\ominus w_{kj} (\beta + r_k^T r_k) + \sum_{i \neq k} w_{ij} r_i^T r_k - r_k^T r_j + \lambda = 0$$

$$w_{kj}^* = \frac{r_k^T r_j - \lambda - \sum_{i \neq k} w_{ij} r_i^T r_k}{r_k^T r_k + \beta}$$